

Architecture & Cost Report

Static NGO Website + CDN

Student Name

Andrew Arthur

Project Title

Hope for Communities NGO Website Hosted on Amazon S3 with Amazon CloudFront

1. Project Overview

This project demonstrates how to host a static website for an NGO using Amazon Web Services (AWS). The website provides information about Hope for Communities NGO, including its mission, programs, gallery, impact, and contact details. The solution uses Amazon S3 for static website hosting and Amazon CloudFront as a Content Delivery Network (CDN) to improve performance and provide secure HTTPS access.

2. Problem Statement

Many NGOs require an affordable, secure, and scalable website to communicate with donors, volunteers, and beneficiaries. Traditional web hosting can be expensive and require ongoing server management. This project provides a low-cost cloud solution using AWS services.

3. AWS Services Used

Amazon S3

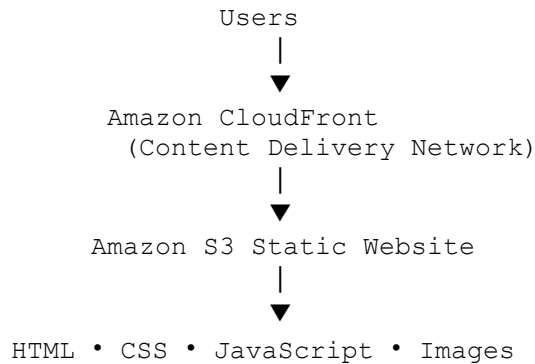
- Hosts the static website files.
- Stores HTML, CSS, JavaScript, and image files.
- Provides reliable and durable storage.

Amazon CloudFront

- Delivers website content through a global CDN.

- Improves loading speed.
 - Provides HTTPS support.
 - Reduces latency for users.
-

4. Architecture



5. Cost Optimization

This project minimizes costs by:

- Using Amazon S3 instead of virtual servers.
- Delivering cached content through CloudFront.
- Eliminating server maintenance.
- Hosting only static files.
- Paying only for storage and bandwidth used.

This architecture is suitable for NGOs because it provides a fast, reliable, and affordable web presence.

6. Challenges Encountered

During the project, several issues were encountered:

- CloudFront initially displayed an older version of the website.
- Some images were broken because the file paths did not match the folder structure.
- The logo size required CSS adjustments.
- Images needed to be moved into an `images` folder to match the HTML references.
- Browser and CloudFront caching required refreshing after updates.

Each issue was resolved by correcting file paths, updating the CSS, organizing the image files, and refreshing the CloudFront cache.

7. Outcome

The completed website includes:

- Responsive homepage
- NGO branding and logo
- About section
- Programs section
- Impact statistics
- Gallery
- Contact information
- Google Maps
- HTTPS access using CloudFront
- Optimized image organization

8. Conclusion

This project demonstrates how AWS cloud services can be used to build a secure, scalable, and cost-effective website for a non-governmental organization. Amazon S3 and CloudFront provide an efficient hosting solution without the need to manage web servers, making this approach ideal for organizations with limited budgets.